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CYBERSECURITY REPORT – LABORATORY 11

1 Problems and questions:

1.1 Are the results obtained by nmap reliable?

Yes, Nmap results are **generally reliable**, but not always exact. **Accuracy depends on network conditions, firewalls, host configuration, and scan options**. Services may be **misidentified** or shown as *open|filtered*, especially with UDP or when **defensive mechanisms are in place**, so results should be validated with additional tools or manual checks.

1.2 Does the information obtained about the target host depend from the scanning options?

Yes. Different scan options significantly change both the **amount and accuracy of information** gathered. **Basic scans may only reveal which ports are open**, while **more advanced options enable deeper analysis**.

For example, **-sS (SYN scan) is faster and stealthier than -sT (TCP connect scan) but may be blocked by firewalls**. The -sV option enables service and version detection, while -O attempts operating system fingerprinting. **Timing options** (-T0 to -T5) affect scan speed and reliability, where **slower scans are more accurate but take longer**. Additionally, NSE scripts (-sC, --script) allow active enumeration and vulnerability detection, revealing web directories, misconfigurations, and known weaknesses that would not appear in a simple port scan.

1.3 Is it permitted to use nmap to scan hosts without permission?

No. Scanning hosts without explicit authorization is generally **illegal and unethical** and may violate laws, organizational policies, or terms of service.

Nmap should only be used on systems you own or have **clear permission** to test (such as personal labs as today or **authorized** penetration tests).

2 Property analysis of available algorithms:

2.1 Run all VM you have installed in VirtualBOX:

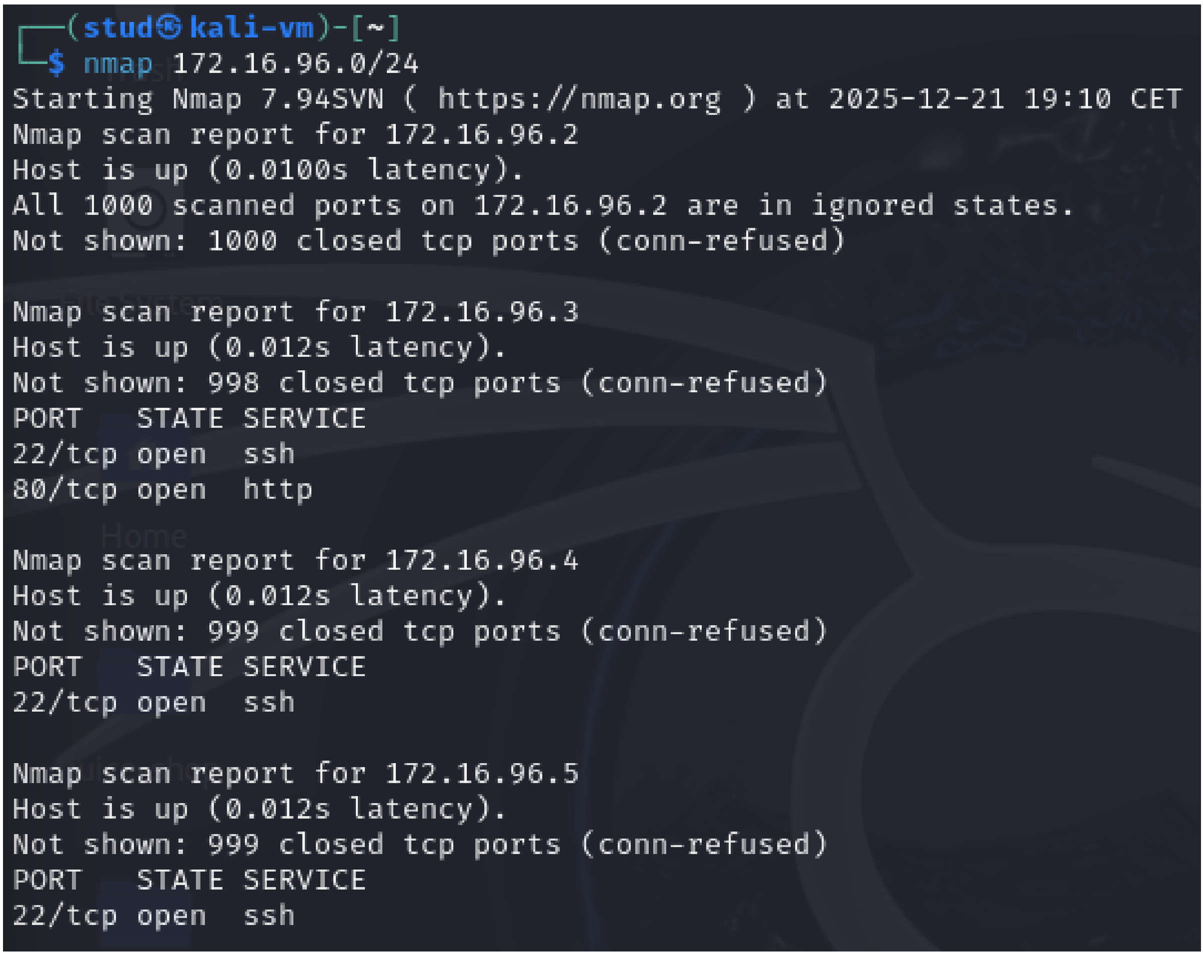
User and password of:

Metasploitable: msfadmin,msfadmin

Kali: stud, stud

Ubuntu: student, student

Execute the following command: > nmap 172.16.96.0/24 (WHAT IS THIS IP ADDRESS AND WHAT IS THE DIFFERNCE WITH THE ADDRESSES BELOW?)



```
┌──(stud㉿kali-vm)-[~]
└─$ nmap 172.16.96.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-21 19:10 CET
Nmap scan report for 172.16.96.2
Host is up (0.0100s latency).
All 1000 scanned ports on 172.16.96.2 are in ignored states.
Not shown: 1000 closed tcp ports (conn-refused)

Nmap scan report for 172.16.96.3
Host is up (0.012s latency).
Not shown: 998 closed tcp ports (conn-refused)
PORT   STATE SERVICE
22/tcp open  ssh
80/tcp open  http

Nmap scan report for 172.16.96.4
Host is up (0.012s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT   STATE SERVICE
22/tcp open  ssh

Nmap scan report for 172.16.96.5
Host is up (0.012s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT   STATE SERVICE
22/tcp open  ssh
```

```
Nmap scan report for 172.16.96.6
Host is up (0.011s latency).
Not shown: 977 closed tcp ports (conn-refused)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown

Nmap done: 256 IP addresses (5 hosts up) scanned in 7.09 seconds
```

172.16.96.0/24 is the network. Scanning it returns which IPs are up and which services/ports they expose. And a single IP **172.16.96.x is a *host*** within that network.

The command shows 5 main things:
- **Active hosts in the subnet:** Out of 256 possible IP addresses, 5 hosts are up: 172.16.96.2, .3, .4, .5, .6.

- **Which TCP ports are open on each host** (top 1000 ports): .3 → SSH (22), HTTP (80), .4 → SSH (22), .5 → SSH (22), .6 → many services (FTP, SSH, Telnet, SMTP, HTTP, SMB, databases, VNC, etc.) and .2 → no open TCP ports found in the default scan.

- **Service identification by port number:** Nmap maps open ports to common services (e.g. 22 → ssh, 80 → http, 3306 → mysql).

- **Port states for scanned ports:** Most ports are closed (conn-refused), also has open ones.

- **Basic network responsiveness** and scan performance: Each host shows low latency (~0.01 s) and the entire /24 network scanned in 7.09 seconds, indicating a local virtual network

2.2 Execute different types of TCP scans and compare results:

A. Options –sT, -sS –sN, -sM, -sA, -sW, -sI.

We have chosen the host of metaexploit as it has many open ports.

nmap <flag> -p 1-1000 172.16.96.6

-sT: Full TCP connect

```
┌──(stud㉿kali-vm)-[~]
└─$ nmap -sT -p 1-1000 172.16.96.6
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-21 19:24 CET
Nmap scan report for 172.16.96.6
Host is up (0.0093s latency).
Not shown: 988 closed tcp ports (conn-refused)
PORT    STATE SERVICE
21/tcp  open  ftp
22/tcp  open  ssh
23/tcp  open  telnet
25/tcp  open  smtp
53/tcp  open  domain
80/tcp  open  http
111/tcp open  rpcbind
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
512/tcp open  exec
513/tcp open  login
514/tcp open  shell

Nmap done: 1 IP address (1 host up) scanned in 0.16 seconds
```

-sS: Half-open SYN (needs sudo)

```
┌──(stud㉿kali-vm)-[~]
└─$ sudo nmap -sS -p 1-1000 172.16.96.6
[sudo] password for stud:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-21 19:24 CET
Nmap scan report for 172.16.96.6
Host is up (0.00075s latency).
Not shown: 988 closed tcp ports (reset)
PORT    STATE SERVICE
21/tcp  open  ftp
22/tcp  open  ssh
23/tcp  open  telnet
25/tcp  open  smtp
53/tcp  open  domain
80/tcp  open  http
111/tcp open  rpcbind
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
512/tcp open  exec
513/tcp open  login
514/tcp open  shell
MAC Address: 08:00:27:A5:A1:BC (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.58 seconds
```

-sN: No TCP flags (needs sudo)

```
┌──(stud㉿kali-vm)-[~]
└─$ sudo nmap -sN -p 1-1000 172.16.96.6

Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-21 19:24 CET
Nmap scan report for 172.16.96.6
Host is up (0.0017s latency).
Not shown: 988 closed tcp ports (reset)
PORT    STATE         SERVICE
21/tcp  open|filtered ftp
22/tcp  open|filtered ssh
23/tcp  open|filtered telnet
25/tcp  open|filtered smtp
53/tcp  open|filtered domain
80/tcp  open|filtered http
111/tcp open|filtered rpcbind
139/tcp open|filtered netbios-ssn
445/tcp open|filtered microsoft-ds
512/tcp open|filtered exec
513/tcp open|filtered login
514/tcp open|filtered shell
MAC Address: 08:00:27:A5:A1:BC (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.73 seconds
```

-sM: FIN/ACK based (needs sudo)

```
┌──(stud㉿kali-vm)-[~]
└─$ sudo nmap -sM -p 1-1000 172.16.96.6
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-22 00:45 CET
Nmap scan report for 172.16.96.6
Host is up (0.0057s latency).
All 1000 scanned ports on 172.16.96.6 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:A5:A1:BC (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.37 seconds
```

-sA: ACK probe (needs sudo)

```
┌──(stud㉿kali-vm)-[~]
└─$ sudo nmap -sA -p 1-1000 172.16.96.6
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-22 00:46 CET
Nmap scan report for 172.16.96.6
Host is up (0.0056s latency).
All 1000 scanned ports on 172.16.96.6 are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)
MAC Address: 08:00:27:A5:A1:BC (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.45 seconds
```

-sW: Window-based ACK (needs sudo)

```
┌──(stud㉿kali-vm)-[~]
└─$ sudo nmap -sW -p 1-1000 172.16.96.6
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-22 00:46 CET
Nmap scan report for 172.16.96.6
Host is up (0.00060s latency).
All 1000 scanned ports on 172.16.96.6 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:A5:A1:BC (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.72 seconds
```

-sI: Idle (spoofed) (needs sudo) (sudo nmap -sI <zombie_IP> 172.16.96.6 -p 80)

Check for possible zombies: > nmap --script ipidseq 172.16.96.0/24

```
┌──(stud㉿kali-vm)-[~]
└─$ nmap --script ipidseq 172.16.96.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-14 00:41 CET
Nmap scan report for 172.16.96.1
Host is up (0.0023s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT     STATE SERVICE
135/tcp  open  msrpc
445/tcp  open  microsoft-ds
2222/tcp open  EtherNetIP-1
MAC Address: 52:55:AC:10:60:01 (Unknown)

Host script results:
|_ipidseq: Unknown

Nmap scan report for 172.16.96.2
Host is up (0.0015s latency).
All 1000 scanned ports on 172.16.96.2 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:11:41:9D (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Host script results:
|_ipidseq: Incremental!
```

```
Nmap scan report for 172.16.96.6
Host is up (0.0036s latency).
Not shown: 976 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
32782/tcp open  unknown
MAC Address: 08:00:27:A5:A1:BC (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Host script results:
|_ipidseq: All zeros

Nmap scan report for 172.16.96.3
Host is up (0.000015s latency).
Not shown: 998 closed tcp ports (reset)
PORT   STATE SERVICE
22/tcp open  ssh
80/tcp open  http

Host script results:
|_ipidseq: All zeros

Nmap done: 256 IP addresses (4 hosts up) scanned in 7.83 seconds
```

Once a zombie is found (_ipidseq: **Incremental**): 172.16.96.2

We can use it as a zombie: > nmap -sI 172.16.96.2 -Pn -p 1-1000 172.16.96.6

```
┌──(stud㉿kali-vm)-[~]
└─$ nmap -sI 172.16.96.2 -Pn -p 1-1000 172.16.96.6
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-14 00:45 CET
Idle scan using zombie 172.16.96.2 (172.16.96.2:80); Class: Incremental
Nmap scan report for 172.16.96.6
Host is up (0.059s latency).
Not shown: 988 closed|filtered tcp ports (no-ipid-change)
PORT    STATE SERVICE
21/tcp  open  ftp
22/tcp  open  ssh
23/tcp  open  telnet
25/tcp  open  smtp
53/tcp  open  domain
80/tcp  open  http
111/tcp open  rpcbind
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
512/tcp open  exec
513/tcp open  login
514/tcp open  shell
MAC Address: 08:00:27:A5:A1:BC (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 12.37 seconds
```

So by using the zombie, Kali's IP is hidden as all the traffic appears as the zombie.

B. Describe the differences between scanning with aforementioned options

The main differences between the different flags are:

- **Accuracy:** -sT and -sS are the most reliable for finding open ports.
- **Firewall/Filter detection:** -sA/-sW are used to detect filtering (filtered vs unfiltered) rather than to find open services.
- **Ambiguous results:** Flag-based scans (-sN, -sM, etc.) often return open|filtered because no response can mean either open (service silently drops) or packets dropped by firewall.
- **Privileges:** Most raw-packet scans (-sS, -sN, -sM, -sA, -sW, -sI) require **root (sudo)**. -sT works without root.
- **Detectability / forensics:** -sT is logged as normal connections; -sS is stealthier but still detectable; -sI hides your IP if successful; -sN/-sM may bypass naive IDS but are less reliable.
- **Environment limits:** Modern OSes and VMs often break -sI by randomizing or zeroing the IP ID (as you observed), making idle scans frequently infeasible.

2.3 Execute UDP scan

We must run: > sudo nmap -sU <target-ip>, all 1000 principal ports are open|filtered

```
┌──(stud㉿kali-vm)-[~]
└─$ sudo nmap -sU --top-ports 25 176.16.96.6
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-22 12:07 CET
Nmap scan report for 176.16.96.6
Host is up (0.00079s latency).

PORT      STATE         SERVICE
53/udp    open|filtered domain
67/udp    open|filtered dhcps
68/udp    open|filtered dhcpc
69/udp    open|filtered tftp
111/udp   open|filtered rpcbind
123/udp   open|filtered ntp
135/udp   open|filtered msrpc
137/udp   open|filtered netbios-ns
138/udp   open|filtered netbios-dgm
139/udp   open|filtered netbios-ssn
161/udp   open|filtered snmp
162/udp   open|filtered snmptrap
445/udp   open|filtered microsoft-ds
500/udp   open|filtered isakmp
514/udp   open|filtered syslog
520/udp   open|filtered route
631/udp   open|filtered ipp
998/udp   open|filtered puparp
1434/udp  open|filtered ms-sql-m
1701/udp  open|filtered L2TP
1900/udp  open|filtered upnp
4500/udp  open|filtered nat-t-ike
5353/udp  open|filtered zeroconf
49152/udp open|filtered unknown
49154/udp open|filtered unknown

Nmap done: 1 IP address (1 host up) scanned in 3.84 seconds
```

```
┌──(stud㉿kali-vm)-[~]
└─$ sudo nmap -sU --top-ports 1000 176.16.96.6
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-22 11:31 CET
Nmap scan report for 176.16.96.6
Host is up (0.0011s latency).
All 1000 scanned ports on 176.16.96.6 are in ignored states.
Not shown: 1000 open|filtered udp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 1622.69 seconds
```

2.4 Try different nmap timing options (-T) from a set of values 0,1,2,3,4,5. What is the difference in results and performance?

> time sudo nmap -sU --top-ports 100 -T0 --reason 176.16.96.6:

```
┌──(stud㉿kali-vm)-[~]
└─$ time sudo nmap -sU --top-ports 100 -T0 --max-retries 1 --min-rate 100 --reason 176.16.96.6
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-22 12:30 CET
Nmap scan report for 176.16.96.6
Host is up, received reset ttl 255 (0.0023s latency).
All 100 scanned ports on 176.16.96.6 are in ignored states.
Not shown: 100 open|filtered udp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 601.35 seconds

real    601.41s
user    0.00s
sys     0.02s
cpu     0%
```

> time sudo nmap -sU --top-ports 100 -T1 --reason 176.16.96.6

```
┌──(stud㉿kali-vm)-[~]
└─$ time sudo nmap -sU --top-ports 100 -T1 --max-retries 1 --min-rate 100 --reason 176.16.96.6
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-22 12:29 CET
Nmap scan report for 176.16.96.6
Host is up, received reset ttl 255 (0.0014s latency).
All 100 scanned ports on 176.16.96.6 are in ignored states.
Not shown: 100 open|filtered udp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 31.18 seconds

real    31.23s
user    0.02s
sys     0.00s
cpu     0%
```

> time sudo nmap -sU --top-ports 100 -T2 --reason 176.16.96.6

```
┌──(stud㉿kali-vm)-[~]
└─$ time sudo nmap -sU --top-ports 100 -T2 --max-retries 1 --min-rate 100 --reason 176.16.96.6
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-22 12:29 CET
Nmap scan report for 176.16.96.6
Host is up, received reset ttl 255 (0.00055s latency).
All 100 scanned ports on 176.16.96.6 are in ignored states.
Not shown: 100 open|filtered udp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 3.09 seconds

real    3.15s
user    0.01s
sys     0.01s
cpu     0%
```

> time sudo nmap -sU --top-ports 100 -T3 --reason 176.16.96.6

```
┌──(stud㉿kali-vm)-[~]
└─$ time sudo nmap -sU --top-ports 100 -T3 --max-retries 1 --min-rate 100 --reason 176.16.96.6
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-22 12:29 CET
Nmap scan report for 176.16.96.6
Host is up, received reset ttl 255 (0.00047s latency).
All 100 scanned ports on 176.16.96.6 are in ignored states.
Not shown: 100 open|filtered udp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 2.34 seconds

real    2.39s
user    0.02s
sys     0.01s
cpu     1%
```

> time sudo nmap -sU --top-ports 100 -T4 --reason 176.16.96.6

```
┌──(stud㉿kali-vm)-[~]
└─$ time sudo nmap -sU --top-ports 100 -T4 --max-retries 1 --min-rate 100 --reason 176.16.96.6
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-22 12:29 CET
Nmap scan report for 176.16.96.6
Host is up, received reset ttl 255 (0.00066s latency).
All 100 scanned ports on 176.16.96.6 are in ignored states.
Not shown: 100 open|filtered udp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 2.38 seconds

real    2.44s
user    0.01s
sys     0.01s
cpu     0%
```

> time sudo nmap -sU --top-ports 100 -T5 --reason 176.16.96.6

```
┌──(stud㉿kali-vm)-[~]
└─$ time sudo nmap -sU --top-ports 100 -T5 --max-retries 1 --min-rate 100 --reason 176.16.96.6
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-22 12:29 CET
Nmap scan report for 176.16.96.6
Host is up, received reset ttl 255 (0.00089s latency).
All 100 scanned ports on 176.16.96.6 are in ignored states.
Not shown: 100 open|filtered udp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 1.33 seconds

real    1.38s
user    0.00s
sys     0.01s
cpu     1%
```

Across all runs the **results were the same**. **Nmap reported all 100 UDP ports as open|filtered (no-response)**. **Performance** varied dramatically: -T5 finished in ~1.4s, -T4/T3 ~2.4s, -T2 ~3.1s, -T1 ~31s and -T0 ~601s. This is because higher -T values use much shorter timeouts, higher parallelism and fewer retries (so Nmap gives up faster), while lower -T values wait longer and retry more (slower but more likely to catch delayed or rate-limited ICMP replies).

open|filtered simply indicates **no response was seen**, not that the port is confirmed open.

2.5 For selected system and selected service use service version detection (-sV) eg. > nmap -sV 172.16.96.x -p 22

```
┌──(stud㉿kali-vm)-[~]
└─$ nmap -sV 172.16.96.6 -p 22
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-22 12:49 CET
Nmap scan report for 172.16.96.6
Host is up (0.0037s latency).

PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.49 seconds
```

The machine is Linux and exposes an old OpenSSH server on port 22.

2.6 Try to discover operating system version installed on VMs (option –O):

What we can see from the sudo nmap -O 172.16.96.6 output

```
┌──(stud㉿kali-vm)-[~]
└─$ sudo nmap -O 172.16.96.6
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-22 12:55 CET
Nmap scan report for 172.16.96.6
Host is up (0.0034s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:A5:A1:BC (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.13 seconds
```

- Many TCP services are open (21, 22, 23, 25, 53, 80, 111, 139, 445, 512–514, 1099, 1524, 2049, 3306, 5432, 5900, 6000, 6667, 8009, 8180, …).
- MAC address 08:00:27:...: **Oracle VirtualBox virtual NIC** (so this is a VM).
- Device type: general purpose and **OS guess: Linux 2.6.X** (detailed range: **2.6.9 - 2.6.33**; CPE shows linux kernel 2.6).
- Network Distance: 1 hop what means that host is on the local network (same subnet/VM host).
- Takeaway: the VM runs many legacy services (telnet, rsh/rlogin, ftp, old DBs, VNC, X11, SMB, NFS, etc.), these are commonly insecure/outdated and good targets for lab testing.

2.7 Check the directory /usr/share/nmap/scripts and find a few various nmap scripts and describe their application:

A. Execute nmap -sC 172.16.96.x (change the IP address)

```
┌──(stud㉿kali-vm)-[~]
└─$ sudo nmap -sC 172.16.96.6
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-22 13:02 CET
Nmap scan report for 172.16.96.6
Host is up (0.0018s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
| ftp-syst:
|   STAT:
| FTP server status:
|      Connected to 172.16.96.3
|      Logged in as ftp
|      TYPE: ASCII
|      No session bandwidth limit
|      Session timeout in seconds is 300
|      Control connection is plain text
|      Data connections will be plain text
|      vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp   open  ssh
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp   open  telnet
25/tcp   open  smtp
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_ssl-date: 2025-12-22T11:04:55+00:00; -58m03s from scanner time.
| sslv2:
|   SSLv2 supported
|   ciphers:
|     SSL2_DES_192_EDE3_CBC_WITH_MD5
|     SSL2_DES_64_CBC_WITH_MD5
|     SSL2_RC4_128_EXPORT40_WITH_MD5
|     SSL2_RC4_128_WITH_MD5
|     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_    SSL2_RC2_128_CBC_WITH_MD5
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
|_Not valid after:  2010-04-16T14:07:45
53/tcp   open  domain
| dns-nsid:
|_  bind.version: 9.4.2
80/tcp   open  http
|_http-title: Metasploitable2 - Linux
111/tcp  open  rpcbind
| rpcinfo:
```

```
|   program version    port/proto  service
|   100000  2            111/tcp   rpcbind
|   100000  2            111/udp   rpcbind
|   100003  2,3,4       2049/tcp   nfs
|   100003  2,3,4       2049/udp   nfs
|   100005  1,2,3      47603/udp   mountd
|   100005  1,2,3      57822/tcp   mountd
|   100021  1,3,4      40902/udp   nlockmgr
|   100021  1,3,4      54047/tcp   nlockmgr
|   100024  1          55535/tcp   status
|_  100024  1          60867/udp   status
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
| mysql-info:
|   Protocol: 10
|   Version: 5.0.51a-3ubuntu5
|   Thread ID: 8
|   Capabilities flags: 43564
|   Some Capabilities: LongColumnFlag, SupportsCompression, SwitchToSSLAfterHandshake, Speaks41ProtocolNew, ConnectWithDatabase, Support41Auth, SupportsTransactions
|   Status: Autocommit
|_  Salt: wT}:E0\_H,@@`@`Et<j%
5432/tcp open  postgresql
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
|_Not valid after:  2010-04-16T14:07:45
|_ssl-date: 2025-12-22T11:04:55+00:00; -58m03s from scanner time.
5900/tcp open  vnc
| vnc-info:
|   Protocol version: 3.3
|   Security types:
|_    VNC Authentication (2)
6000/tcp open  X11
6667/tcp open  irc
| irc-info:
|   users: 1
|   servers: 1
|   lusers: 1
|   lservers: 0
|   server: irc.Metasploitable.LAN
```

```
|   version: Unreal3.2.8.1. irc.Metasploitable.LAN
|   uptime: 0 days, 11:32:03
|   source ident: nmap
|   source host: D6594DEC.70F3DAAE.168799A3.IP
|_  error: Closing Link: gpdunzrag[172.16.96.3] (Quit: gpdunzrag)
8009/tcp open  ajp13
|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open  unknown
|_http-title: Apache Tomcat/5.5
|_http-favicon: Apache Tomcat
MAC Address: 08:00:27:A5:A1:BC (Oracle VirtualBox virtual NIC)

Host script results:
|_smb2-time: Protocol negotiation failed (SMB2)
|_clock-skew: mean: 16m57s, deviation: 2h30m01s, median: -58m03s
| smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_  System time: 2025-12-22T06:04:27-05:00
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)

Nmap done: 1 IP address (1 host up) scanned in 75.82 seconds
```

B. Execute nmap --script http-enum,http-headers,http-methods,http-phpversion

Running: > sudo nmap -p 80,8180 -sV --script=http-enum,http-headers,http-methods,http-php-version --reason 172.16.96.6

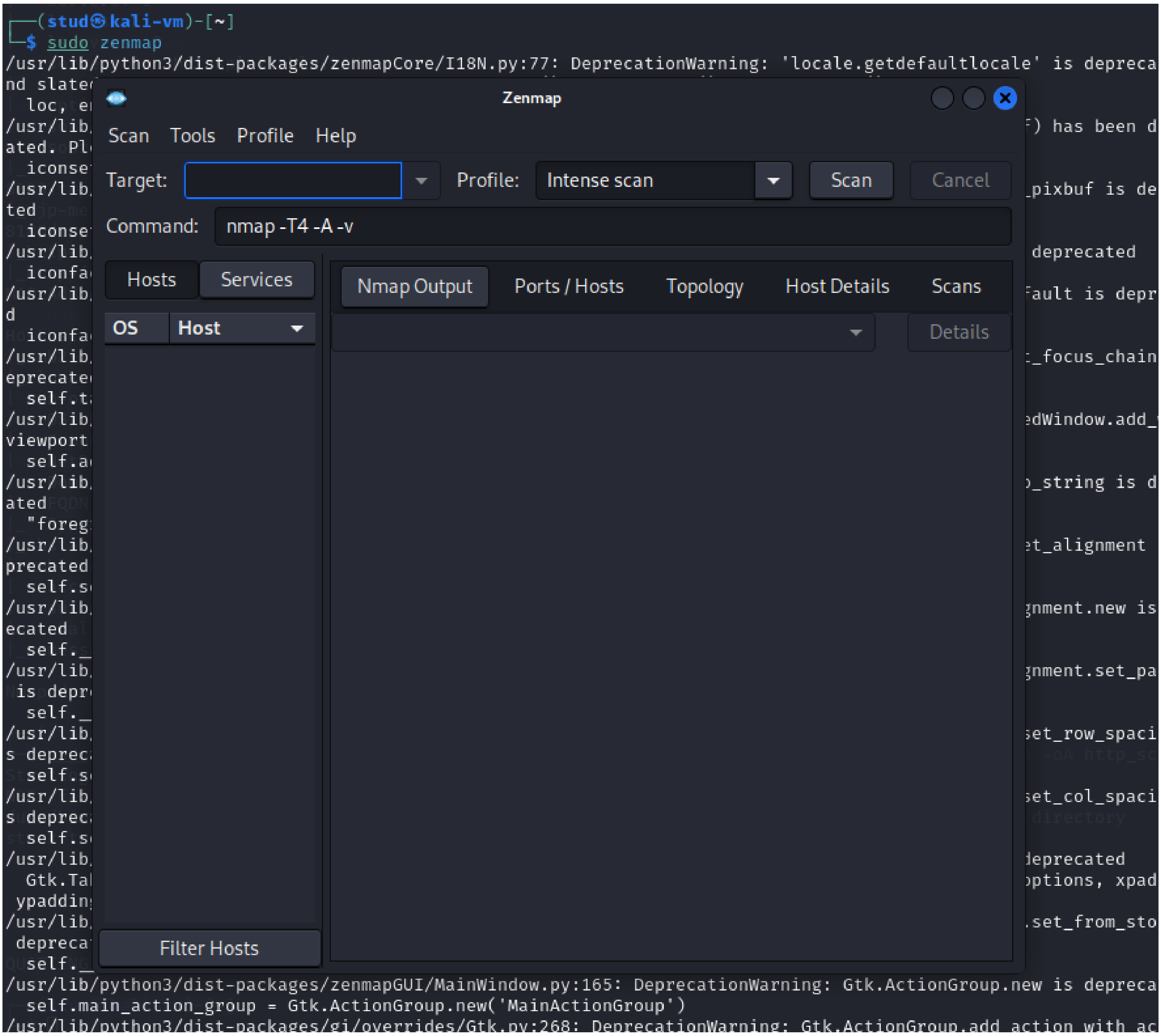
```
┌──(stud㉿kali-vm)-[~]
└─$ sudo nmap -p 80,8180 -sV --script=http-enum,http-headers,http-methods,http-php-version --reason 172.16.96.6
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-12-22 13:13 CET
Nmap scan report for 172.16.96.6
Host is up, received arp-response (0.0047s latency).

PORT     STATE SERVICE REASON         VERSION
80/tcp   open  http    syn-ack ttl 64 Apache httpd 2.2.8 ((Ubuntu) DAV/2)
| http-headers:
|   Date: Mon, 22 Dec 2025 11:16:08 GMT
|   Server: Apache/2.2.8 (Ubuntu) DAV/2
|   X-Powered-By: PHP/5.2.4-2ubuntu5.10
|   Connection: close
|   Content-Type: text/html
|
|_  (Request type: HEAD)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
| http-methods:
|_  Supported Methods: GET HEAD POST OPTIONS
| http-php-version: Versions from logo query (less accurate): 5.1.3 - 5.1.6, 5.2.0 - 5.2.17
| Versions from credits query (more accurate): 5.2.3 - 5.2.5, 5.2.6RC3
|_Version from header x-powered-by: PHP/5.2.4-2ubuntu5.10
| http-enum:
|   /tikiwiki/: Tikiwiki
|   /test/: Test page
|   /phpinfo.php: Possible information file
|   /phpMyAdmin/: phpMyAdmin
|   /doc/: Potentially interesting directory w/ listing on 'apache/2.2.8 (ubuntu) dav/2'
|   /icons/: Potentially interesting folder w/ directory listing
|_  /index/: Potentially interesting folder
8180/tcp open  http    syn-ack ttl 64 Apache Tomcat/Coyote JSP engine 1.1
| http-headers:
|   Server: Apache-Coyote/1.1
|   Content-Type: text/html;charset=ISO-8859-1
|   Date: Mon, 22 Dec 2025 11:16:08 GMT
|   Connection: close
|
|_  (Request type: HEAD)
|_http-server-header: Apache-Coyote/1.1
| http-methods:
|_  Supported Methods: GET HEAD POST OPTIONS
MAC Address: 08:00:27:A5:A1:BC (Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 35.95 seconds
```

Scans show two HTTP services and several clear weaknesses: **port 80** is Apache **2.2.8 (Ubuntu)** serving PHP **5.2.4-2ubuntu5.10** (old, EOL and likely vulnerable); http-enum found **/phpMyAdmin**, **/phpinfo.php**, **/tikiwiki/** and directories with likely listings (/doc, /icons), phpinfo.php can leak config and phpMyAdmin is a high-value attack surface. http-methods shows only GET, HEAD, POST, OPTIONS (no risky PUT/DELETE), but the server and PHP versions are the main risk. **Port 8180** runs **Apache Tomcat/Coyote**

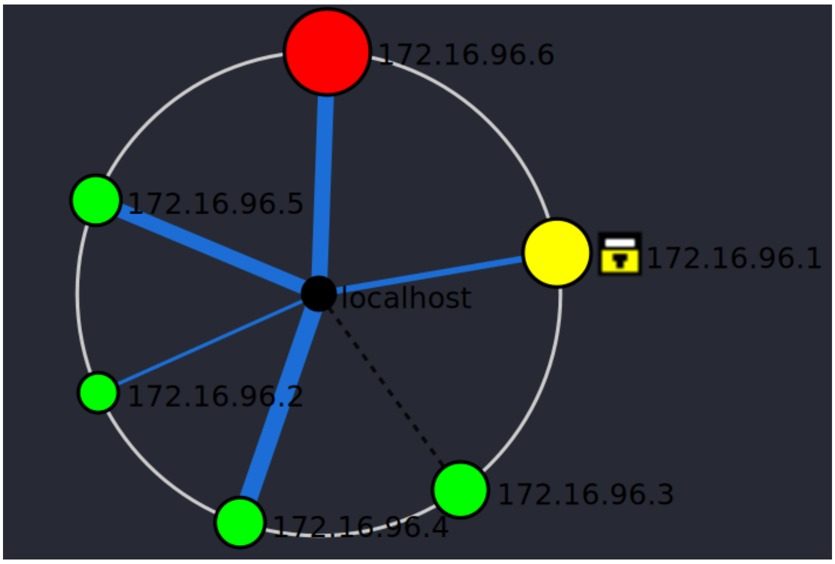
1.1 (also outdated). Combined with the earlier full-host scan (telnet, ftp anonymous, old MySQL/Postgres, VNC, rsh/login services, Linux 2.6 kernel, VirtualBox MAC), this VM is intentionally vulnerable and a good target for lab exploitation.

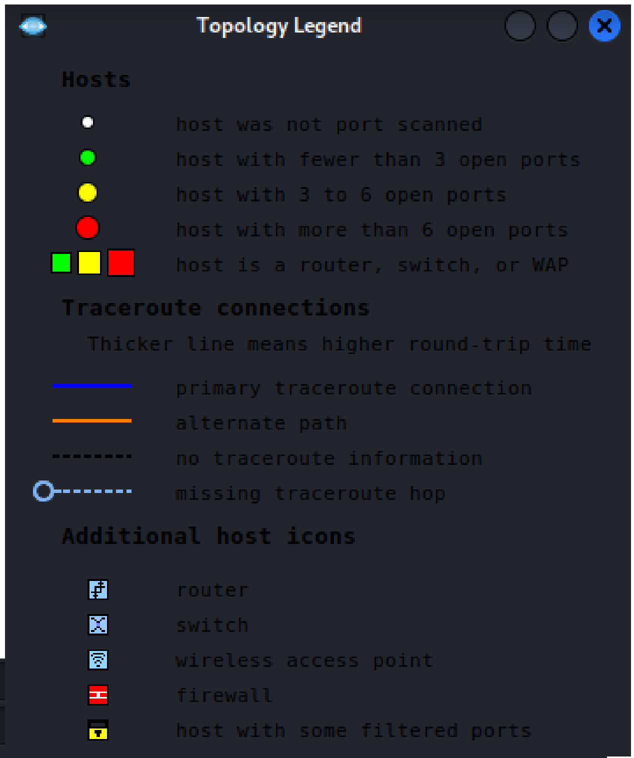
2.8 Use Zenmap to discover the Virtual Network structure and VMs operating systems and active services

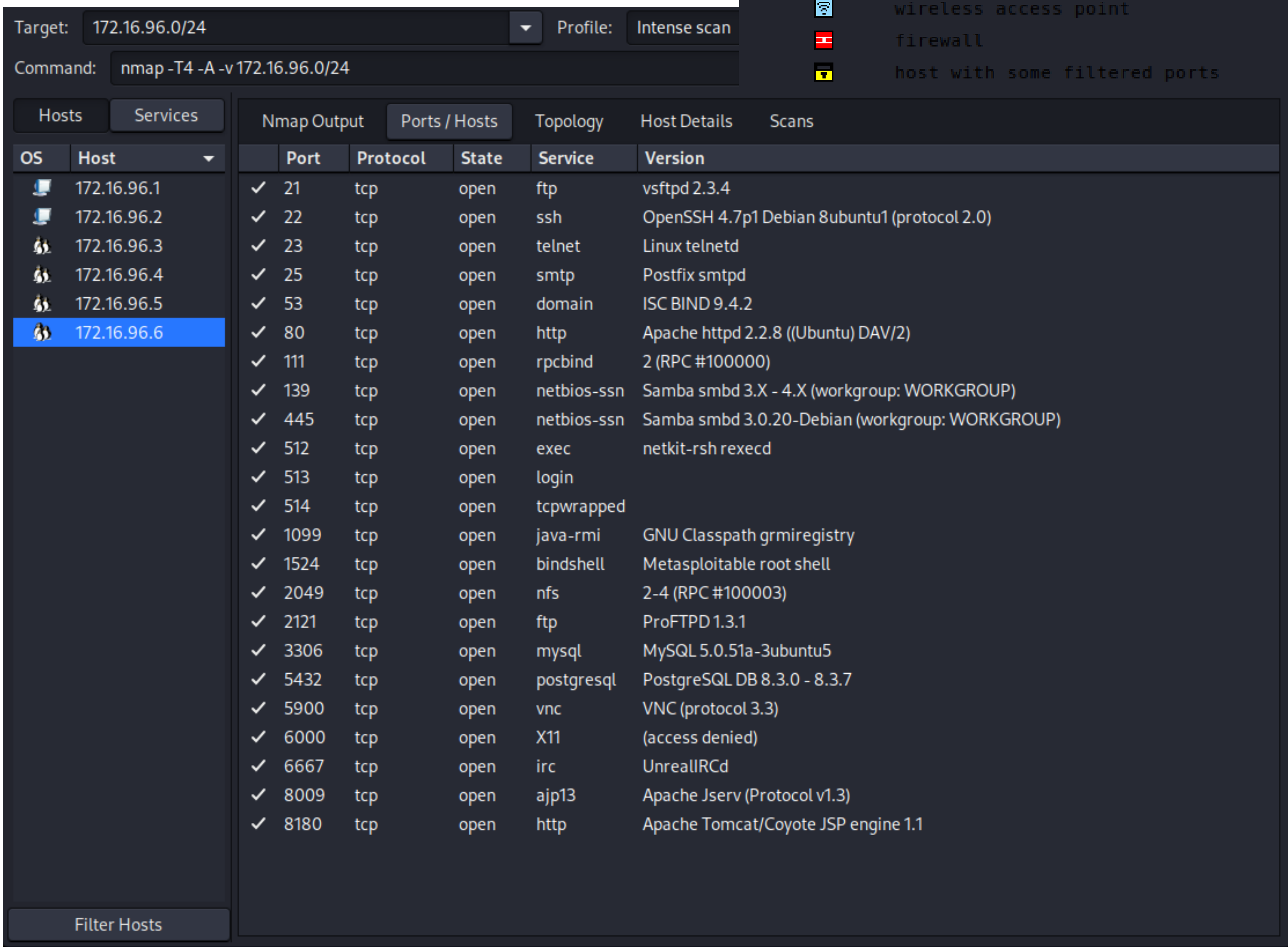
By installing and running: > sudo zenmap



By running the scan with target: > 172.16.96.0/24







	Port	Protocol	State	Service	Version
✓	21	tcp	open	ftp	vsftpd 2.3.4
✓	22	tcp	open	ssh	OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
✓	23	tcp	open	telnet	Linux telnetd
✓	25	tcp	open	smtp	Postfix smtpd
✓	53	tcp	open	domain	ISC BIND 9.4.2
✓	80	tcp	open	http	Apache httpd 2.2.8 ((Ubuntu) DAV/2)
✓	111	tcp	open	rpcbind	2 (RPC #100000)
✓	139	tcp	open	netbios-ssn	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
✓	445	tcp	open	netbios-ssn	Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
✓	512	tcp	open	exec	netkit-rsh rexecd
✓	513	tcp	open	login	
✓	514	tcp	open	tcpwrapped	
✓	1099	tcp	open	java-rmi	GNU Classpath grmiregistry
✓	1524	tcp	open	bindshell	Metasploitable root shell
✓	2049	tcp	open	nfs	2-4 (RPC #100003)
✓	2121	tcp	open	ftp	ProFTPD 1.3.1
✓	3306	tcp	open	mysql	MySQL 5.0.51a-3ubuntu5
✓	5432	tcp	open	postgresql	PostgreSQL DB 8.3.0 - 8.3.7
✓	5900	tcp	open	vnc	VNC (protocol 3.3)
✓	6000	tcp	open	X11	(access denied)
✓	6667	tcp	open	irc	UnrealIRCd
✓	8009	tcp	open	ajp13	Apache Jserv (Protocol v1.3)
✓	8180	tcp	open	http	Apache Tomcat/Coyote JSP engine 1.1

You can check all the ports open per device in the network in a visual way.

2.9 Use amap tool against selected VM and compare the results with those obtained during nmap scan > amap -bq 172.16.96.x 80 3306

amap v5.4 (www.thc.org/thc-amap) started at 2025-12-22 15:04:06 - APPLICATION MAPPING mode

Protocol on 172.16.96.6:3306/tcp matches mysql - banner: >\n5.0.51a-3ubuntu5da9;Z3/2,Nph=4.({W">U

Protocol on 172.16.96.6:80/tcp matches http - banner: <html><head><title>Metasploitable2 - Linux</title></head><body>\n<pre>\n\n _ _ _ _ _ ____ \n _ __ ___ ___| |_ __ _ ___ _ __ | | ___ (_) |_ __ _| |__ | | ______ \\ \n| '_ ` _ \\/ _ \\__/ _`/ _

Protocol on 172.16.96.6:80/tcp matches http-apache-2 - banner: HTTP/1.1 200 OK\r\nDate Mon, 22 Dec 2025 130603 GMT\r\nServer Apache/2..8 (Ubuntu) DAV/2\r\nX-Powered-By PHP/5.2.4-2ubuntu5.10\r\nConnection close\r\nContent-Type text/html\r\n\r\n<html><head><title>Metaspoitable2 - Linux</title></head><body>\n<pre>\n\n

amap v5.4 finished at 2025-12-22 15:04:12

Nmap previously detected **port 80** as **Apache 2.2.8 (Ubuntu) with PHP 5.2.4** and **port 3306** as **MySQL**, using service/version fingerprinting (-sV). Both tools agree on the services and versions, but **amap provides raw protocol banners and application-level fingerprints**, while **nmap provides structured service/version detection and OS context**. This shows that amap complements nmap by confirming services through banner analysis, whereas nmap gives broader and more precise reconnaissance results.